



Assignment 2

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1. Executive Summary

This proposal serves as a review of the cyber-security needs of Maybank, Malaysia's largest financial institution that has come under scrutiny following allegations of customer data leaks. Although Maybank has publicly confirmed that it has not experienced any breaches to date and that its systems remain secure, these allegations demonstrate the increasing cyber risk that financial institutions around the world face [1]. A bank manages millions of consumer accounts, processes countless financial transactions and holds vast amounts of personal data and, therefore, should have a strong cyber-security program to preserve both the integrity of its systems and the confidence of its customers.

After reviewing the advantages and disadvantages of each of the above approaches, it is recommended that Maybank develop a hybrid approach that would augment their internal resources by utilizing some aspects of external resources. This hybrid, multi-pronged approach would provide Maybank with improved protection, enhanced regulatory compliance and greater preparedness for impending threats.

2. Background

a. Background of the Company

Maybank Financial Group is the largest banking group in Malaysia, as well as one of Southeast Asia's leading financial-services companies. With over six million customers using their digital platforms such as Maybank2u, and MAE mobile apps, Maybank offers access to all types of online banking, payment, transfer, and financial services [2].

Cyber-crime is a big threat with Maybank. The sheer size of their customer base, and the sensitive nature of their business, lead them to have high-value targets for cyber-pirates wanting to steal money or sensitive data.

b. Problem Statement

In July 2024, an alleged threat actor posted to a dark-web forum that they had uploaded or leaked a database purported to belong to Maybank2u. The site also indicated that they were selling customers' personal information, login credentials, and other sensitive documents [3]. Maybank responded by assuring the public that all systems were secure and that internal investigations did not confirm any breach, but there was widespread fear

regarding this incident [4]. This scenario illustrates how false claims about leaked data affect customer confidence, damage reputations, and require a large amount of investigative resources.

Additionally, Maybank has previously failed to comply with Bank Negara Malaysia's regulations regarding system downtime due to inadequate operational resilience and was subsequently fined RM 4.32 million by them [5]. As a result of this action, they will now be subject to the regulatory penalties due to the impact of technological. Subsequently, Maybank has put in place new restrictions to prohibit access to their Online Bank from old devices or browsers, thus indicating that Cyber-Security Threats continue to change and progress [6].

This combination of factors highlights an urgent requirement for Maybank and its sub-banks to enhance their Cyber-Security Infrastructure, improve their Incident Response capabilities, and develop better Internal Monitoring Processes. The Cyber-Security Risks are not only external but include risks internally posed by Insider Threats, Phishing and Credential Misuse, and System Configurations Errors, thus creating a challenging environment for Digital Banking. As a result of the rapidly evolving abilities of Cyber-Criminals, they need to adopt an Integrated Security Strategy that will be able to withstand future risks.

c. Setting up an Internal Security Team

a. Identify Project Team

To protect Maybank’s data and operations comprehensively, I propose establishing an internal cyber-security project team. Below is a suggested team structure, and stakeholders.

Table 1: Project Team

Role	Responsibility / Duties
Chief Information Security Officer (CISO) / Head of Security	Overall leadership of security strategy, governance, policy compliance, regulatory oversight, reporting to senior management.
Project Manager (Security Implementation Lead)	Plan and oversee implementation of security initiatives (infrastructure upgrades, monitoring tools, processes), schedule & resource management, coordination across departments.

Security / Network Analyst(s)	Day-to-day monitoring of network traffic, vulnerability scanning, patch management, configuration management, log analysis, compliance checks.
Incident Response Team (IRT)	Detect, triage, and respond to security incidents (data leaks, unauthorized access, fraud attempts); perform root cause analysis; coordinate remediation and reporting.
Compliance & Risk Officer	Ensure compliance with regulatory and data protection laws (e.g. PDPA, relevant banking/financial regulations), conduct audits, manage risk assessments, document policies and procedures.
Security Awareness & Training Coordinator	Develop and deliver staff training on security practices (phishing awareness, secure credentials, data handling), maintain policies and documentation, enforce access control and least-privilege practices.

b. Identify Stakeholders

Table 2: Stakeholders

Stakeholder	Role / Interest in the Project
Senior Management / Board of Directors	For funding approval, oversight, strategic alignment.
IT Department / Infrastructure Team	Integration of security controls with existing infrastructure; cooperation on patching, backups, system configuration.
Customer Data & Operations Teams	Handling client data, ensuring business processes remain secure.
Compliance & Legal Department	To ensure regulatory compliance and manage legal risks.
Customers / Clients	Indirectly stakeholders whose data is protected; their trust and confidence matter.
External Auditors / Regulators	May review security posture, compliance with Financial Services regulations, data protection laws.

3.Proposal

a. Option 1: Do nothing

Maintaining the current state of the security infrastructure without significant changes is the first alternative. There will be no upfront cost to maintain the current setup, but there is an extremely high risk associated with this alternative. If there were ever to be a real cyber-attack, Maybank would incur a significant loss due to the financial impact, fines imposed by regulators, and irreversible harm to their reputation.

Additionally, maintaining the current state of the security infrastructure puts Maybank at risk for phishing attacks, insider threats, and more sophisticated cyber-attacks that could compromise customer accounts and their financial systems. The current level of protection offered by Maybank is simply not a sustainable option given the size and scope of their business.

b. Option 2: Build a Cyber security team

The second alternative is to create a complete in-house cybersecurity unit. Having full control over your data, systems, and cybersecurity protocols is one of the major benefits of establishing such a unit. It would allow for the customized development of Maybank's security solutions based upon its current technology and organizational processes. Internal staff already know the systems, which decreases the amount of time required to respond to incidents; therefore, the time from detection through to recovery will be reduced.

Establishing an internal team will also provide an increase in institutional knowledge and provide Maybank less reliance upon external vendor solutions. However, developing such a team requires a tremendous amount of resources in terms of salary, training, and retaining highly skilled cybersecurity professionals.

The organization would also need to provide 24/7 monitoring, which can be difficult to maintain with a fully internal staff. Furthermore, the limited number of available cybersecurity professionals in the marketplace makes hiring costly and time-consuming.

c. Option 3: Outsourcing to an External company

The third solution for a bank's cyber-security is utilizing the services of an outside cyber-security provider. Using an outside provider allows access to advanced security technology, as well as to an expert level of knowledge concerning cyber-security. In addition to the

availability of 24-Hour Security Operations Centre (SOC) monitoring, vendors frequently provide real-time threat intelligence and keep logs of their experience in dealing with the most sophisticated cyber-attacks. In most instances, outsourcing results in decreased requirements of internal staff and may be less expensive initially.

There are also risks associated with outsourcing that should be evaluated. The transfer of sensitive customer information to a vendor is one of these risks. Vendor lock-in may result from long-term contracts. The degree to which an organisation has control over the monitoring and responding to incidents will be reduced with an outsourcing model.

The bank must also ensure that the outsourced services provided are in compliance with applicable legislation and that any failures on the part of the vendor will negatively affect the reputation of Maybank.

4. Risk Assessment

a. Risk Assessment

Table 3: Risk Matrix Diagram

		Probability				
Impact		5	4	3	2	1
	5	Very High	Very High	Very High	High	High
	4	Very High	High	High	Medium	Medium
	3	High	High	Medium	Medium	Low
	2	High	Medium	Medium	Low	Very Low
	1	Medium	Low	Low	Very Low	Very Low

This section will focus on the probability of risk and impact. This will be determined using a scale factor of 1 to 5 and is measured from very high to low.

Table 4: Risk & impact table

Risks	Consequence	Probability	Impact	Risk Level
Do Nothing	Hackers may continue exploiting vulnerabilities, gaining unauthorized access to customer banking data such as account	1	5	High

	numbers, phone numbers, and emails. • Maybank may suffer severe financial losses due to fraud compensation and operational disruptions. • Customer trust will deteriorate, damaging Maybank's brand reputation. • Regulatory non-compliance could result in penalties from Bank Negara Malaysia under RMiT and PDPA laws.			
Build Internal Security Team	• Shortage of skilled cybersecurity professionals could delay incident response and system hardening. • Internal staff may require extensive training, increasing cost and implementation time. • If not properly managed, insider threats may continue to pose risks, including misuse of access privileges.	4	2	Medium
Outsource to External Cybersecurity Vendor	• Dependence on external vendors may introduce third-party security risks. • A breach in the vendor's systems can expose Maybank's sensitive data.	3	4	High

	• Misalignment in service expectations may lead to delayed response times during critical cyber incidents.			
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b. ROSI

1. Annual Loss Expectancy (ALE)

$$ALE = SLE \times ARO$$

Assumptions for Maybank's latest breach scenario

- Asset Value (AV) = RM 12,000,000
(Customer data + reputational impact + operational disruption)
- Exposure Factor (EF) = 0.25
(25% data compromised in an event)
- Single Loss Expectancy (SLE):

$$SLE = 12,000,000 \times 0.25 = \text{RM } 3,000,000$$

- Annual Rate of Occurrence (ARO): = 1
(One significant breach per year is realistic for banks)

Therefore:

$$ALE = 3,000,000 \times 1 = \text{RM } 3,000,000 \text{ per year}$$

2. Mitigation Ratio

Mitigation ratio describes how much the proposed security solution reduces the risk.

For Internal Security Team, Estimated reduction in breach likelihood is 60%.

$$\text{Mitigation Ratio} = 0.60$$

For Outsourcing to Vendor, Estimated reduction in breach likelihood is 75%

$$\text{Mitigation Ratio} = 0.75$$

These are standard industry values used in financial-sector risk assessments.

3. Cost of Solution

Table 5: Build Internal Security Team

Item	Estimated Annual Cost
Cybersecurity staff salaries (5 personnel)	RM 900,000
Training & certifications	RM 150,000
SIEM system + threat monitoring tools	RM 450,000
Incident response & penetration testing	RM 200,000
Total Annual Investment	RM 1,700,000

Table 6: Outsourcing to External Vendor

Item	Estimated Annual Cost
Managed SOC (24/7 monitoring)	RM 1,200,000
Threat intelligence services	RM 250,000
Incident response retainer	RM 300,000
Compliance monitoring	RM 150,000
Total Annual Investment	RM 1,900,000

4. Return on Security Investment (ROSI)

$$\text{ROSI} = \frac{(\text{ALE} \times \text{Mitigation Ratio}) - \text{Cost of Solution}}{\text{Cost of Solution}}$$

ROSI for Internal Security Team

$$\begin{aligned}\text{ROSI} &= \frac{(3,000,000 \times 0.60) - 1,700,000}{1,700,000} \\ &= \frac{1,800,000 - 1,700,000}{1,700,000} \\ &= 0.0588 = 5.9\%\end{aligned}$$

A positive ROSI indicates the internal team provides financial value, though modest.

ROSI for Outsourcing

$$\begin{aligned}\text{ROSI} &= \frac{(3,000,000 \times 0.75) - 1,900,000}{1,900,000} \\ &= \frac{2,250,000 - 1,900,000}{1,900,000} \\ &= \frac{350,000}{1,900,000} \\ &= 0.1842 = 18.4\%\end{aligned}$$

Outsourcing generates much higher ROI because it reduces more risk despite higher cost.

Table 7: Overall Review

Calculation	Internal Team	Outsourced Vendor
ALE	RM 3,000,000	RM 3,000,000
Mitigation Ratio	0.60	0.75
Annual Cost	RM 1,700,000	RM 1,900,000
ROSI	5.9%	18.4%
Conclusion	Small but positive ROI	Best financial return

c. The Significance of Addressing the Issue

It is critical to address Maybank's Cybersecurity concerns in the context of compliance with government regulations, as well as How cyber threats can impact an organisation's long-

term viability, credibility and efficiency. Cybersecurity has shifted from being a single discipline within the financial services sector to becoming part of an organisation-wide effort to protect an organisation's reputation and provide confidence to Customers about the protection of their Personal Information, in addition to providing the economy of the Country With a higher level of Security. Cyber threats are becoming increasingly sophisticated; therefore, organisations who ignore them will be exposed to significant Financial losses as well as damage to their Reputation, which may take multiple generations to rebuild. In order to improve their overall Strategic Position (resiliency) and meet the needs of the millions of people in Malaysia who depend on their services, Maybank must take Proactive Steps to protect itself against Cybersecurity threats.

National brands rely on consumer confidence to thrive and meet expectations. In the banking sector, national banks operate with customers' money and provide access to their accounts and all financial transactions. Consequently, the concern is that any potential breach of security could cause a customer to question whether or not their funds or sensitive personal data were safe and secure with their chosen financial institution. If customers do not have confidence in the safety of their accounts, they will withdraw their deposits, close their accounts, and potentially change banks, resulting in an ongoing loss of projected revenue over time. Therefore, by improving cybersecurity, Maybank demonstrates to customers that they are making an effort to protect their data and maintain their trust, especially during this period when most of what their customers conduct on a daily basis is conducted via the internet. Additionally, it is important to note that building a strong level of security is more than just technical in nature; it is a strategic strategy that helps to build loyalty among existing customers, while also protecting the reputation of their brand.

Another significant aspect of managing risks associated with cyber threats is compliance with laws and regulations. The financial services sector in Malaysia is highly regulated by Bank Negara Malaysia (the central bank of Malaysia) as a result of various regulatory frameworks in place, including the RMIT (Risk Management in Technology), PDPA (Personal Data Protection Act) and Cybersecurity Guidelines from Bank Negara Malaysia. A failure to protect the organisation's systems properly may lead to heavy penalties, increased regulatory oversight, and mandatory audits that disrupt the organisation's operations. Proactively mitigating cyber risks allows Maybank to mitigate unnecessary fines and demonstrates the bank's good corporate governance, which increases

confidence in the bank among its stakeholders (regulators, investors, industry partners, etc.) and enhances the organisation's ability to operate smoothly with reduced compliance risk.

In addition, addressing the challenges associated with cybersecurity enables Maybank to minimise losses associated with fraud, including fraud, breaches of personal data, system downtime, and external investigations. An example of this would be a major breach that could cost millions in compensation to customers impacted by the breach, as well as legal costs, forensic investigation costs, system recovery costs, and monitoring of fraud by customers. When the costs of implementing robust cybersecurity measures are compared to the costs of protecting systems from the threat of cybercrime, the clarity of the financial benefits of investing in prevention becomes clearer. Therefore, by investing in internal security personnel or outsourcing to a third party, Maybank will materially decrease the risk and severity of future fraud, thus, justifying the financial investment in cybersecurity based on long-term financial savings.

In conclusion, resolving Cybersecurity Issues faced by Maybank helps to maintain customer confidence and trust, meet regulators' requirements, minimise losses caused by Cybercrime and protect brand reputation as well as positioning for Digital Competitiveness in Banking today. Cyber Security should not just be considered as a technical obligation, it is an important function to protect the whole organisation in terms of its future viability.

5. Financial Plan

a. Budget Allocation Estimation

Proposed budget for the first year is RM 3.25 million, and the annual recurring will be lower which is approximately RM 1.5 million.

Table 8: Budget Allocation

Budget Category	Estimated Cost (RM)	Description of Use
Security Tools & Technology	RM 1,000,000	Purchase SIEM (Security Information & Event Management), endpoint protection, firewall upgrades, MFA, and network monitoring tools.

Cybersecurity Team Salaries	RM 1,500,000	Hiring 1 CSO, 2 security analysts, 1 incident responder, 1 compliance officer. Includes training and certifications.
Outsourced Managed Security Services (MSSP)	RM 350,000	24/7 monitoring, threat intelligence, external audits, and fast incident response.
Training & Awareness Programs	RM 150,000	Staff cybersecurity training, phishing simulation programs, and awareness campaigns.
Infrastructure Upgrades	RM 150,000	Server hardening, secure cloud integration, encryption upgrades.
Emergency & Contingency Fund	RM 100,000	For zero-day vulnerabilities, emergency forensics, or recovery.

Maybank's allocated budget is primarily for strengthening the bank's cybersecurity posture to eliminate the possibility of another incident like recently occurred, but part of the budget will be used to upgrade outdated security tools and monitoring capabilities and investing in the latest advanced technologies such as firewalls, SIEM systems, endpoint protection platforms and multi-factor authentication [7]. This includes enhancing the bank's ability to detect, prevent and respond to cyber threats in real time as well as investing heavily in the establishment of an internal cybersecurity team, including the hiring of a Chief Security Officer, analysts and incident response specialists so as to establish professionals who are dedicated to the ongoing monitoring and protection of the bank's systems.

The budget will also provide funds to hire out certain functions to a reliable managed security service provider (MSSP) that provides 24/7 monitoring, more rapid threat detection, and expert-level incident response. These services are extremely important for a continually operating financial institution. The budget allocation will also be used to provide training and awareness courses to teach Maybank workers about cybersecurity threats as well as to help them identify phishing or social engineering scams. A portion of the budget will provide for infrastructure upgrades, like secure cloud storage, enhancements to encryption, and hardening of network security. The last portion of the budget will be set aside for contingencies in order

to allow the bank to be able to react immediately in the case of any unexpected cyber incidents, zero-day exploits, or emergency investigations.

b. Key Activities

- Implement advanced security monitoring systems (SIEM, EDR, MFA).
- Recruit and train an internal cybersecurity team.
- Integrate external 24/7 threat monitoring from MSSP.
- Conduct penetration testing, vulnerability assessments, and regular audits.
- Upgrade firewalls, servers, and encryption mechanisms.
- Launch cybersecurity awareness and phishing simulation programs for staff.
- Maintain backup and disaster recovery improvements.
- Reserve emergency funds for immediate incident response.

6. Conclusion

To sum up, careful planning, clear communication and involvement of internal and external stakeholders will determine if this project will be successful. The allocation of a realistic budget, coordination of resources, and making sure all stakeholders know what their responsibilities are will enable the smooth and efficient delivery of this project. The planned activities, such as, staffing, marketing, logistics, and technology support will provide for the operational needs of the project within its budget. Thus, if managed and collaborated properly, the project will be in a good position to meet its goals and produce a positive impact.

7. References

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